

BSC121 Revision Question Set 2025/09

UNIT 1

1. Evaluate $\lim_{x \rightarrow 0} \frac{|2x-1| - |2x+1|}{2x}$ or show that it does not exist.
2. Evaluate $\lim_{x \rightarrow \infty} e^{-2x} \cos(x)$.
3. Given the function, $f(x) = \begin{cases} x+2, & \text{if } x < 2 \\ ax^2, & \text{if } x = 2 \\ b, & \text{if } x > 2 \end{cases}$. Find the value of a and b that will make f continuous at $x = 2$.

UNIT 2

1. Find the derivative of the following functions.
 - (a) $f(x) = -2^x + \tan x$
 - (b) $g(x) = x^2 \sin x$
 - (c) $h(x) = e^{1-2x^2}$
 - (d) $k(x) = \sin(x) + x^2 \tan^{-1}(x)$
 - (e) $m(x) = (\sin x)^{\ln x}$
2. Find $y'(x)$ for the implicit function, $y^2 e^x = \ln(x^2 + y^2)$.
3. Determine the absolute extreme values for $f(x) = 3x(x+4)^{\frac{2}{3}}$ on the interval $[-5, -1]$.
4. Use L'Hopital's Rule to evaluate the limit, $\lim_{x \rightarrow 1^+} \left[(x-1) \tan\left(\frac{\pi}{2}x\right) \right]$.

UNIT 3

1. Solve the following indefinite integral.

(a) $\int \sqrt[3]{x} + 10\sqrt[5]{x^3} dx$

(b) $\int x^3 - \frac{e^{-x} - 4}{e^{-x}} dx$

(c) $\int \frac{6}{x^3} - \frac{2}{x} dx$

2. Solve the following indefinite integral.

(a) $\int (8x - 12)(4x^2 - 12x)^4 dx$

(b) $\int \sec(1 - x) \tan(1 - x) dx$

(c) $\int \left(\frac{15}{x^2} - 5x \right) \cos \left(\frac{6}{x} + x^2 \right) dx$

3. Differentiate each of the following integrals with respect to x .

(a) $\int_4^x 9 \cos^2(t^2 - 6t + 1) dt$

(b) $\int_7^{\sin(6x)} \sqrt{t^2 + 4} dt$

(c) $\int_x^{x^2} e^{t^2} dt$

4. Evaluate the following integrals, if possible. If it is not possible, clearly explain why it is not possible to evaluate the integral.

(a) $\int_0^1 3(4x + x^4)(10x^2 + x^5 - 2)^6 dx$

(b) $\int_0^{\frac{\pi}{4}} \frac{8 \cos(2x)}{\sqrt{9 - 5 \sin(2x)}} dx$

5. Determine the area of the region bounded by the given set of curves.

(a) $y = x^2 + 2, y = \sin(x), x = -1$ and $x = 2$.

(b) $y = \frac{8}{x}, y = 2x$ and $x = 4$.

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6. Use the method of disks/washers to determine the volume of the solid obtained by rotating the region by the given curves about the given line.

- (a) Rotate the region bounded by $y = \sqrt{x}$, $y = 3$ and the y -axis about the y -axis.
- (b) $x = 0$ and $x = 1$ about the line $y = -2$.
- (c) Rotate the region bounded by $y = 10 - 6x + x^2$ and $y = -10 + 6x - x^2$, $x = 1$ and $x = 5$ about the line $y = 8$.

7. Use the method of cylinders to determine the volume of the solid obtained by rotating the region by the given curves about the given line.

- (a) Rotate the region bounded by $y = \frac{1}{x}$, $x = \frac{1}{2}$, $x = 4$ and the x -axis about the y -axis.
- (b) Rotate the region bounded by $y = 2x + 1$, $y = 3$ and $x = 4$ about the line $y = 10$.

8. Evaluate each of the following integrals.

- (a) $\int 4x \cos(2 - 3x) dx$
- (b) $\int_6^0 (2 + 5x) e^{\frac{1}{3}x} dx$
- (c) $\int \sin^3\left(\frac{2}{3}x\right) \cos^4\left(\frac{2}{3}x\right) dx$
- (d) $\int \cos^4(2x) dx$
- (e) $\int \sin(8x) \cos(3x) dx$
- (f) $\int_1^3 \sin(8x) \sin(x) dx$

9. Use a trigo-substitution to evaluate each of the following integrals.

- (a) $\int \frac{\sqrt{x^2 + 16}}{x^4} dx$
- (b) $\int \sqrt{1 - 7x^2} dx$

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10. Evaluate each of the following integrals.

(a) $\int \frac{4}{x^2 + 5x - 14} dx$

(b) $\int_{-1}^0 \frac{x^2 + 7x}{(x+2)(x-1)(x-4)} dx$

(c) $\int \frac{8}{3x^3 + 7x^2 + 4x} dx$

(d) $\int_2^4 \frac{3x^2 + 1}{(x+1)(x-5)^2} dx$

11. Determine if each of the following integrals converge or diverge. If the integral converges, determine its value.

(a) $\int_0^{\infty} (1+2x)e^{-x} dx$

(b) $\int_1^2 \frac{4x}{\sqrt[3]{x^2 - 4}} dx$

12. Set up, but do not evaluate, an integral for the length $y = \sqrt{x+2}, 1 \leq x \leq 7$ using

(a) $ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx,$

(b) $ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy.$

13. Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $x = \sqrt{y+5}, \sqrt{5} \leq x \leq 3$ about y -axis using

(a) $ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

(b) $ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

UNIT 4

1. Determine if the following series is convergent or divergent.

(a) $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

(b) $\sum_{n=0}^{\infty} n e^{-n^2}$

(c) $\sum_{n=1}^{\infty} \frac{n}{n^2 - \cos^2 n}$

(d) $\sum_{n=0}^{\infty} \frac{1}{3^n - n}$

(e) $\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{n^2 + 5}$

2. Determine the interval of convergence for the following power series.

(a) $\sum_{n=1}^{\infty} \frac{2^n}{n} (4x - 8)^n$

(b) $\sum_{n=0}^{\infty} n! (2x + 1)^n$

(c) $\sum_{n=1}^{\infty} \frac{x^{2n}}{(-3)^n}$

3. Find a power series representation for the function $g(x) = \frac{1}{1+x^3}$ and determine its interval of convergence.

4. By using the Maclaurin's series for $\ln(1+x)$, or otherwise, find the first four terms in the series expansion of $\ln\left(\frac{1+3x}{1-2x}\right)$. Hence, use the expansion to show that

$$\ln\left(\frac{7}{2}\right) \approx \frac{3725}{3072}.$$

UNIT 5

1. Solve the following first order differentiate equation.

(a) $y'e^{-y} = (2x - 4)$

(b) $\frac{dr}{d\theta} = \frac{r^2}{\theta}$

(c) $2y' - y = 4\sin 3t$

(d) $\frac{dv}{dt} = 10 - 2v$

(e) $\frac{dx}{dt} = 5x - 3$

2. Solve the following IVP.

(a) $y' = \frac{xy^3}{\sqrt{1+x^2}}; y(0) = -1$

(b) $y' = \frac{3x^2 + 4x - 4}{2y - 4}; y(1) = 3$

(c) $ty' + 2y = t^2 - t + 1; y(1) = \frac{1}{2}$

(d) $y'' + 11y' + 24y = 0; y(0) = 0, y'(0) = -7$

(e) $y'' - 4y' + 9y = 0; y(0) = 0, y'(0) = -8$

(f) $y'' - 4y' + 4y = 0; y(0) = 12, y'(0) = -3$